

Public Parking Management System

Simple Workshop Project Requirements

1. Project Overview

A small web-based system for managing a government or local-authority public roadside parking area. Parking employees can record vehicles entering the parking area, issue a simple parking ticket/receipt, and record vehicle exits. The system automatically calculates the parking fee based on configurable rates and provides a simple end-of-day income and outstanding-payment summary.

2. User Roles

Administrator	Manage parking fee rates, employees, and view reports.
Parking Officer	Record vehicle entry/exit, collect payments, and print parking tickets/receipts.

3. Login

Employees log in using a username/email and password. The system displays features according to the user's role.

4. Vehicle Entry

When a vehicle arrives, the parking officer records:

- Vehicle number
- Vehicle type (optional)
- Entry date and time (automatically recorded)
- Parking officer (automatically linked to the logged-in user)

After saving the entry, the system generates a parking ticket/receipt that can be printed and placed on the vehicle.

5. Vehicle Exit & Payment

When a vehicle leaves, the officer enters or searches the vehicle number. The system finds the active parking session, calculates the duration, applies the configured pricing rules, and displays the amount to be paid. After payment is recorded, the session is marked as completed and a receipt can be printed.

6. Configurable Parking Rates

The administrator can configure the pricing rules instead of hard-coding them. For example:

Parking Duration	Example Rate
First hour	Rs. 120
Second hour	Rs. 150
Each additional hour	Rs. 150

The actual rates are examples and should be configurable by the administrator.

7. Dashboard

The main dashboard should provide a quick view of:

- Currently parked vehicles
- Today's vehicle entries
- Today's completed/paid vehicles

- Today's total income
- Outstanding/unpaid parking sessions

8. Daily Report

At the end of the day, an authorized user can view a simple report containing total vehicles, paid/completed sessions, unpaid or outstanding sessions, and total collected income. The report can optionally be printed or exported.

9. Suggested Small Additions

- Search parking history by vehicle number.
- Show the current parking duration for active vehicles.
- Allow cancellation/voiding of an incorrect parking entry with a reason.
- Keep a basic payment method field such as Cash/Card.
- Record the officer who handled each entry and payment for accountability.

10. Core Screens

1	Login
2	Dashboard
3	Vehicle Entry / New Parking
4	Print Parking Ticket
5	Vehicle Exit / Payment
6	Parking Rate Settings
7	Parking History
8	Daily Report
9	Employee Management (Admin)

11. Out of Scope

To keep the workshop project small, the system does not require online payment gateways, mobile applications, GPS/parking-slot tracking, customer accounts, advanced analytics, or integration with external government systems.

12. Basic Workflow

Login → Vehicle Arrives → Record Vehicle Number & Entry Time → Print Ticket → Vehicle Leaves → Search Vehicle Number → Calculate Duration & Fee → Record Payment → Print Receipt → Daily Report